

Rogers-Ramanujan Derivation — cycles 16-18

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Contents

Rogers-Ramanujan Derivation — cycles 16-18	1
Introduction	1
Definitions and Notation	1
Results	2
Cycle Chronology	2
Examples and Regression Checks	3
Remarks	3
Open Questions	3
References	3
Appendix: Implementation Details	4

Rogers-Ramanujan Derivation — cycles 16-18

Introduction

Cycles 16-18 were closure cycles. They did not add new proof steps, scripts, data files, figures, ledger events, or documentation. Their purpose was to confirm that the previously completed Rogers-Ramanujan derivation remained closed and that no concrete formal defect had been identified.

The directive was to derive the two Rogers-Ramanujan identities in a formal power series setting, without web search or external lookup, and to separate experimental computation from proof. That directive had already been completed before these cycles. The validated route remains:

1. Work in the formal power series ring $\mathbb{Z}[[q]]$.
2. Derive finite q -binomial and Jacobi-type product identities internally.
3. Derive a triangular Bailey-style matrix transform rather than cite it.
4. Use coefficientwise limiting to connect the Rogers-Ramanujan series to the Jacobi specializations.
5. Use formal unit cancellation and modulo-5 product factorization to obtain the two product sides.

The audit report supplied for cycles 16-18 records no critical or moderate findings. It reports only standing minor organizational warnings from `promise_check` and `org_check`, and concludes **VALIDATED**.

Definitions and Notation

The proof campaign uses $\mathbb{Z}[[q]]$, the ring of formal power series with integer coefficients. In this setting, equality means coefficientwise equality: two series are equal when every coefficient of q^n agrees.

The finite q -Pochhammer symbol is

$$(q; q)_n = \prod_{k=1}^n (1 - q^k).$$

Infinite products are interpreted formally through coefficientwise stabilization: for any fixed coefficient degree, only finitely many product factors can affect that coefficient.

A unit is a formal power series with constant term 1 or -1 . Such a series has a multiplicative inverse in $\mathbb{Z}[[q]]$. This is the algebraic basis for the product cancellations used in the final proof.

Results

Theorem 1: First Rogers-Ramanujan Identity

The validated final theorem L41 states:

$$\sum_{n \geq 0} \frac{q^{n^2}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+1})(1 - q^{5m+4})}.$$

Cycles 16-18 did not alter this theorem. They report it as already proved in `docs/proof/final_proof.md` and validated through milestone M4.

Theorem 2: Second Rogers-Ramanujan Identity

The validated final theorem L42 states:

$$\sum_{n \geq 0} \frac{q^{n^2+n}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+2})(1 - q^{5m+3})}.$$

Cycles 16-18 also leave this theorem unchanged. The cycle records repeatedly identify `docs/proof/final_proof.md`, `docs/lemmas/lemma_catalogue.md`, `docs/validation.md`, and `promise_ledger.jsonl` as the durable completion artifacts.

Proof Chain Status

The proof chain remained unchanged across all three cycles:

Component	Status
M0 formal setup	Validated
M1 finite approximant exploration	Validated
M2 transfer mechanism	Validated
M3 product-side formalization	Validated
M4 final proof synthesis and validation separation	Validated
Primary directive	Complete

The accepted mechanism is the same in all cycle records: finite q -binomial algebra gives the Jacobi-type product identity; the internally derived triangular transform links the Rogers-Ramanujan series to those Jacobi specializations; coefficientwise limiting and unit cancellation in $\mathbb{Z}[[q]]$ yield the residue-class product identities.

Cycle Chronology

Cycle 16

The cycle 16 researcher session `8fbe8907-953f-4fa9-8068-ebe760429210` classified the run as final closure. It listed no open questions and recommended no build or run.

The cycle 16 worker session `15a8dcfc-f63d-4041-b5df-01fc97687ff4` built nothing and ran no commands. It stated that all milestones M0-M4 were already validated.

The cycle 16 auditor session `4209740f-e746-40b5-b1d6-c189ddd1470d` returned **VALIDATED**. It reported no critical or moderate findings, no file modifications, and only standing minor warnings from the validator tools.

Cycle 17

The cycle 17 researcher session `0c56fc38-152b-435c-b355-b4c55970c75f` again identified the run as final closure. It stated that no worker cycle should be started unless a specific formal defect is found.

The cycle 17 worker session `c9c03af6-16bc-4b4f-8f2c-69d44908c8bf` built no artifact and ran no commands. It repeated that no scripts, figures, data files, or ledger events were needed.

The cycle 17 auditor session `e914f012-8bc6-4783-83f0-62783b566eae` returned **VALIDATED**. It found no concrete algebraic or formal defect and confirmed that the worker output introduced no new proof obligations.

Cycle 18

The cycle 18 researcher session `d49c6be0-64c9-4155-96a0-f207bd64f9f4` recorded final closure and no open mathematical or organizational questions.

The cycle 18 worker session `f5394a72-b655-49bb-b989-9e356f4f76d4` built nothing and ran no commands. It identified the proof as complete in $\mathbb{Z}[[q]]$.

The cycle 18 auditor session `779f783e-aea1-4d0f-910e-eb1a69160f34` returned **VALIDATED**. It reported no proof defect and no file changes. Its rationale names the same completed proof mechanism: q -binomial/Jacobi algebra, triangular transform, coefficientwise limiting, and product cancellation.

Examples and Regression Checks

No new examples or regression checks were produced in cycles 16-18.

The standing regression record remains in `docs/validation.md` and the existing data under `data/finite_experiments/`. These computations are classified as discovery or regression support, not as substitutes for proof. The durable checks include Bailey-transform residuals, product-side residuals, $KMAX = 0$ edge checks, finite approximant comparisons, and diagnostic records for rejected routes.

No figures are embedded in this report because no new figures were produced during cycles 16-18.

Remarks

The closure decision has a narrow condition for reopening: a specific formal defect would need to be identified in one of the proof-bearing steps, such as a coefficientwise limit, exponent specialization, triangular transform derivation, Jacobi specialization, unit cancellation, or modulo-5 residue cancellation.

The reports do not recommend continuing the primary proof search. Optional future work, such as seeking a direct partition bijection, would be a new objective rather than a requirement for completing the current directive.

Open Questions

There are no open questions for the primary Rogers-Ramanujan directive.

The standing nonblocking warnings are organizational:

Warning	Status
<code>promise_check</code> orphan-artifact warnings for session database files and generated cycle reports	Standing minor warning; command exits 0
<code>org_check</code> root-file warning for <code>rogers_ramanujan_run_config.yaml</code>	Standing minor warning; command exits 0
Missing <code>REFERENCES.md</code>	Expected; no external references were used

References

No external references are cited in this report. No `REFERENCES.md` file exists in the workspace.

Appendix: Implementation Details

Durable Artifacts

Artifact	Role
plan_of_record.md	Defines the directive, goals, milestones M0-M4, and success criteria
docs/proof/final_proof.md	Self-contained final derivation of both identities
docs/proof/bailey_matrix_transform.md	Proof-bearing transfer mechanism for M2
docs/proof/product_side.md	Formal modulo-5 product cancellation for M3
docs/lemmas/lemma_catalogue.md	Lemma catalogue through final theorem entries L41-L42
docs/validation.md	Separates proof-bearing results from discovery and regression checks
promise_ledger.jsonl	Ledger of milestone events and validation decisions
MANIFEST.md	Current project inventory and cross-reference map

Current File Counts

Item	Count or lines
MANIFEST.md	60 lines
plan_of_record.md	113 lines
docs/proof/final_proof.md	270 lines
docs/proof/product_side.md	140 lines
docs/proof/bailey_matrix_transform.md	270 lines
docs/lemmas/lemma_catalogue.md	810 lines
docs/validation.md	382 lines
promise_ledger.jsonl	33 events
Wolfram scripts under scripts/rr	10 scripts, 1405 lines
Python scripts under scripts/rr	10 scripts, 1166 lines
Total scripts under scripts/rr	20 scripts, 2571 lines
Proof, lemma, and validation docs	13 files, 3036 lines
Experiment data and figure files under data/finite_experiments	82 files

Script Inventory

Script	Lines	Purpose
scripts/rr/bailey_matrix_probe.wls	142	Checks the derived triangular matrix transform, alpha solves, pair residuals, and limiting inner-sum residuals
scripts/rr/euler_tail_telescoping_probe.wls	155	Searches exact low-order Euler-tail divergence certificates and records finite truncation consistency
scripts/rr/finite_gap_probe.wls	71	Tests largest-part gap-two recurrences and naive bounded residue-product comparisons
scripts/rr/finite_rr_experiments.wls	106	Generates truncated series/product coefficients, differences, recurrence probes, and negative controls

Script	Lines	Purpose
scripts/rr/mod5_state_recurrence_probe.wls	167	Wolfram probe for shifted diagonal and modulo-5 state recurrence diagnostics
scripts/rr/nonlocal_involution_probe.wls	172	Enumerates nonlocal subset-transfer candidates and classifies involution failures
scripts/rr/product_side_formal_check.wls	99	Checks transformed, Jacobi, and final reciprocal residue-product residuals
scripts/rr/product_transform_probe.wls	190	Tests Euler-multiplied finite series, finite Jacobi identities, candidate transformed finite formulas, and negative controls
scripts/rr/q_difference_probe.wls	123	Checks series-side q-difference ladder residuals and product/backsolve diagnostics
scripts/rr/transformed_cancellation_probe.wls	180	Enumerates signed transformed-series objects and tests local absorb/release cancellation
scripts/rr/direct_bijection_probe.py	409	Enumerates finite gap-two and residue partitions, tests beta/abacus signatures, and renders abacus diagnostics
scripts/rr/mod5_state_recurrence_probe.py	234	Python mirror for modulo-5 diagnostics and stable coefficient tables
scripts/rr/plot_bailey_transform_residuals.py	49	Renders residual checks for the Bailey-style triangular transform
scripts/rr/plot_euler_tail_lattice_residuals.py	80	Renders Euler-tail lattice support and finite residual diagnostics
scripts/rr/plot_mod5_state_support.py	71	Renders stable coefficient support by residue class
scripts/rr/plot_nonlocal_involution_fixed_points.py	65	Renders nonlocal fixed/unpaired object patterns
scripts/rr/plot_product_transform_residuals.py	82	Renders first-failure-degree residual plot for product-transform candidates
scripts/rr/plot_q_difference_residuals.py	72	Renders residual magnitude plot for q-difference diagnostics
scripts/rr/plot_rr_difference_heatmap.py	50	Renders heatmap of coefficient differences for target and negative-control finite products
scripts/rr/plot_transformed_cancellation_survivors.py	54	Renders transformed signed-survivor pattern and boundary corrections

Figure Inventory

The current figure files are:

Figure	Status in this report
data/finite_experiments/bailey_transform_residuals.png	Existing figure; not new in cycles 16-18

Figure	Status in this report
data/finite_experiments/direct_bijection_abacus_examples.png	Existing figure; not new in cycles 16-18
data/finite_experiments/euler_tail_lattice_residuals.png	Existing figure; not new in cycles 16-18
data/finite_experiments/mod5_state_support.png	Existing figure; not new in cycles 16-18
data/finite_experiments/nonlocal_involution_fixed_points.png	Existing figure; not new in cycles 16-18
data/finite_experiments/product_transform_residuals.png	Existing figure; not new in cycles 16-18
data/finite_experiments/q_difference_residuals.png	Existing figure; not new in cycles 16-18
data/finite_experiments/rr_difference_heatmap.png	Existing figure; not new in cycles 16-18
data/finite_experiments/transformed_cancellation_fixings.png	Existing figure; not new in cycles 16-18

Session Cross-Reference

Cycle	Role	Session ID	Contents
16	Researcher	8f8e8907-953f-4fa9-8068-e5a760429210	brief; no open questions; no further work recommended
16	Worker	15a8dcfc-f63d-4041-b5df-016d97887aff	no commands; all milestones validated
16	Auditor	4209740f-e746-40b5-b1d6-c189bd11470d	no critical or moderate findings; no files changed
17	Researcher	0c56fc38-152b-435c-b355-b4b55970a75f	reopen only on concrete formal defect
17	Worker	c9c03af6-16bc-4b4f-8f2c-65d44902c8b1	built; no commands run; completion artifacts identified
17	Auditor	e914f012-8bc6-4783-83f0-6273bd5552ae	no new claims or proof obligations
18	Researcher	d49c6be0-64c9-4155-96a0-f207b1d44f9f	no mathematical or organizational question open
18	Worker	f5394a72-b655-49bb-b989-9835664ff76d	no defect; no commands; proof complete in $\mathbb{Z}[[q]]$
18	Auditor	779f783e-aea1-4d0f-910e-e11a59416d34	no proof defect; no files changed

Cross-Reference Map

Source	Consuming record	Role
docs/lemmas/lemma_catalogue.md L0-L42	docs/proof/final_proof.md	Supplies the formal setup, transfer lemmas, product lemmas, and final theorem statements
docs/proof/bailey_matrix_transform.md	docs/proof/final_proof.md	Provides the validated M2 transfer mechanism
docs/proof/product_side.md	docs/proof/final_proof.md	Provides the validated M3 modulo-5 product cancellation
docs/proof/final_proof.md	promise_ledger.jsonl, cycle 16-18 sessions	Durable final proof artifact for M4

Source	Consuming record	Role
<code>docs/validation.md</code>	cycle 16-18 audit reports	Records that computation is discovery/regression support, not proof
<code>MANIFEST.md</code>	this appendix	Supplies current script inventory, cumulative stats, and artifact map

Source Inventory

The source material used for this report was:

Source	Date or state	How it fits the timeline
Provided audit report for cycles 16-18	Current input	Establishes the closure decision, minor warnings, and no-file-change status
<code>data/sessions.db</code> session records	2026-05-15	Supplies the nine cycle 16-18 researcher, worker, and auditor records
<code>plan_of_record.md</code>	Existing durable plan	Defines milestones M0-M4 and the directive success criteria
<code>MANIFEST.md</code>	Current 60-line snapshot	Supplies implementation inventory and cross-references
<code>promise_ledger.jsonl</code>	33 events	Records milestone validation history through M4
<code>docs/validation.md</code>	Current validation document	Separates proof steps from computations and rejected routes
<code>docs/proof/final_proof.md</code>	Current final proof	Durable proof artifact for both identities
Workspace file inventory	Current filesystem state	Confirms no <code>REFERENCES.md</code> and lists current reports, scripts, docs, data, and figures

Cycles 16-18 contain no source gap that affects the report. The only absences are intentional: no new worker artifacts, no new computations, and no external reference file.